Discrete Consensus-Based Optimization

Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory

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Outline

1. Introduction
2. Emergence of Consensus and Global Optimization
3. Numerical Simulations
4. Conclusion

Goal

- **Goal.** For a closed subset $S \subset \mathbb{R}^d$ and $f : S \to \mathbb{R}$, we address

$$\min_{x \in S} f(x),$$

where f is (Borel) measurable on S and can be evaluated at any point.

- Our method is motivated by *consensus-based optimization* (CBO).
  - CBO is a metaheuristic methods described by a system of stochastic differential equations (SDEs).
  - CBO utilizes spatial dynamics among agents instead of convexity or derivative information.
  - CBO enable the construction of proofs of convergence to a global minimum.

- *Metaheuristic* is generic algorithm framework that...
  - find a good enough solution in a reasonable time.
  - it can be applied to almost all optimization problems.

- Let X_t^i be a position of i-th agent (particle) at time t. Then CBO on $\mathbb{R}^d$ is

$$dX_t^i = \lambda F(\bar{X}_t - X_t^i)dt + \sigma G(\bar{X}_t - X_t^i)dW_t^i. \tag{CBO}$$

- W_t^i is a d-dimensional Brownian motion at time t assigned to the i-th agent.
- Functions F and G are designed to be zero at the origin.
- Typically, $\bar{X}$ is defined by the "softmin" operator

$$\bar{X}_t = \frac{\sum_{i=1}^N X_t^i w_f^\beta(X_t^i)}{\sum_{k=1}^N w_f^\beta(X_t^i)}, \quad w_f^\beta(X) := e^{-\beta f(X)}, \quad \beta > 0,$$

  which is a weighted average of particles $X_t^1, \dots, X_t^N$.
- The smallest value of f get the largest weight. This softmin action gets precise when $\beta \to \infty$; $\bar{X}_t$ converges to the centroid of X_t^i's which minimizes f.
- Therefore, the updates terminates when every agent arrives at a common point; this phenomenon is referred to as a *consensus*.

- Convergence of CBO is justified by its stochastic mean-field limit.

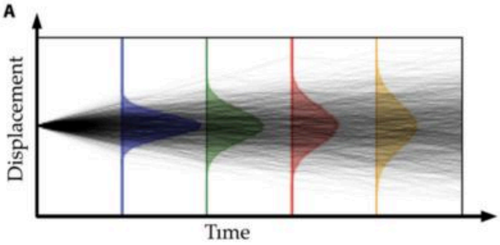

Figure: SDEs (Brownian motion) approximated by PDE (heat equation).[1]

- In the case of CBO, for $F(x) = x$, $G(x) = \sqrt{2}|x| I_d$, one has

$$\begin{aligned} &\partial_t \rho_t = \mathrm{div}(\mu[\rho_t]\rho_t) + \Delta(\kappa[\rho_t]\rho_t), \quad \mu[\rho_t](x) : -\lambda(v_f^\beta[\rho_t] - x), \\ &\kappa[\rho_t](x) := \sigma^2 \| v_f^\beta[\rho_t] - x\|^2, \quad v_f^\beta[\rho_t] = \frac{\int_{\mathbb{R}^d} x w_f^\beta(x) d\rho_t(x)}{\int_{\mathbb{R}^d} w_f^\beta(x) d\rho_t(x)}, \end{aligned} \tag{PCBO}$$

where ρ_t is a law of the agents at time t.

[1] Rudy, S. H., Brunton, S. L., Proctor, J. L., & Kutz, J. N. (2017). Data-driven discovery of partial differential equations. Science advances, 3(4), e1602614.

- Mean-field analysis provides a substantial insight into the operation of CBO.

- Nevertheless, it is challenging to determine how far the results from the mean-field model apply to the actual workhorse.

- Primarily, the mean-field model assumes that the initial distribution includes a global minimizer.
  - This inherently presupposes knowledge of its location!

- The clarity on to what extent implemented CBO can perform global optimization also remains uncertain.

- This motivates us to introduce the 'discrete' model in which mathematical analysis is feasible.

Proposition

Suppose that f has a unique minimizer x^* and appropriate conditions hold. If the initial data is well-prepared and $x_* \in \mathrm{supp}(\rho_0)$, then for any sufficiently small $\varepsilon > 0$, there exists $\beta_0 > 0$ such that $\beta > \beta_0$ implies the following assertions.

1. Let W_2 denotes the Wasserstein-2 distance. There exists some positive constants $A, B > 0$ such that

$$W_2^2(\rho_t, \delta_{x^*}) \le Ae^{-B(2\lambda - d\sigma^2)t},$$

until some time horizon $T > 0$ that $W_2^2(\rho_t, \delta_{x^*})$ becomes ε at T.

2. Let $((X^i_{k\Delta t})_{k=0,\dots,K})_{i\in[N]}$ be a iterations generated by the Euler-Maruyama discretization scheme from (PCBO) with timestep Δt, where K matches the time horizon $K\Delta t = T$. Then we have

$$\left\| \frac{1}{N} \sum_{i=1}^{N} X^i_{K\Delta t} - x^* \right\|^2 \le \varepsilon_{\mathrm{total}},$$

with probability larger than $1 - (\delta + \varepsilon_{\mathrm{total}}^{-1}(C_{\mathrm{NA}}\Delta t + C_{\mathrm{MFA}} N^{-1} + 6\varepsilon))$, where $\delta, \varepsilon, C_{\mathrm{NA}}, C_{\mathrm{MFA}}$ are suitable positive constants, and C_{NA} depends linearly on dimension d and the number of agents N.

Discrete Consensus-Based Optimization

- We consider two random dynamical systems defined as

$$F_1(x,y,\eta) = x + \gamma^1(y-x) + \gamma^2(y-x)\odot\eta, \quad \text{or}$$
$$F_2(x,y,\eta) = x + \bar{\gamma}^1(y-x) + \bar{\gamma}^2\|y-x\|\frac{\eta}{\sqrt{d}}. \tag{Model}$$

- For a nonempty closed set S, we apply the following update rule:

$$\begin{cases} x_{n+1}^i = F^i(x_n^i, p_n, \eta_n^i), \quad n \in \mathbb{N}\cup\{0\}, \quad i \in \{1,2,\dots,N\} =: [N], \\ p_n := x_n^{\min \mathcal{I}_n}, \quad \mathcal{I}_n := \{i \in [N] \mid f(x_n^i) = \min_{j\in[N]} f(x_n^j)\}, \\ x_n^i\big|_{n=0} = x_0^i \in S, \quad F^i = F_1 \text{ or } F_2. \end{cases} \tag{DCBO}$$

- We assume that S is the effective domain of f:

$$f : \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}, \quad f(x) = +\infty \iff x \notin S,$$

- If S is a convex set such that the projection onto S, say $\mathcal{P}_S$, is defined, then

$$x_{n+1}^i := \mathcal{P}_S(F(x_n^i, p_n, \eta_n^i)).$$

DCBO Algorithm

Algorithm 2.1 DCBO

Input: function f, closed domain S, parameters (γ^1, γ^2) or $(\bar{\gamma}^1, \bar{\gamma}^2)$, the number of agents N, stopping criterions max_iter and max_dist
Initialize x^i in S for all $i \in [N]$
$p \leftarrow x^{\min \mathcal{I}}, \quad \text{where} \quad \mathcal{I} = \{i \in [N] \mid f(x^i) = \min_{j \in [N]} f(x^j)\}$
$n \leftarrow 0$
while $n <$ max_iter and $\max \|x^i - x^j\| \geq$ max_dist **do**
    $x^i \leftarrow F^i(x^i, p, \eta_n^i)$ for all $i \in [N]$
    if Projection $\mathcal{P}_S$ into S is defined **then**
        $x^i \leftarrow \mathcal{P}_S(x^i)$ for all $i \in [N]$
    end if
    Update $p \leftarrow x^{\min \mathcal{I}}, \quad \text{where} \quad \mathcal{I} = \{i \in [N] \mid f(x^i) = \min_{j \in [N]} f(x^j)\}$
    $n \leftarrow n + 1$
end while
return $(p, f(p))$.

Remark on the Modeling Spirit

- Seamless use of p_n is attributed to the fact that (DCBO) is a *discretized*.
- The construction of a well-defined solution to a continuous SDE for p_t is not standard.
- At the PDE level, defining p_t is even less clear; the concept of 'i-th index' does not exists anymore.
- DCBO phenomenologically mirrors CBO: In (Model), F_1, F_2 can be formally derived from the previous models by $\beta \to \infty$.
- Thus, DCBO is inherently discrete CBO-type algorithm and not a *discretized* version of CBO.

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- The updates terminates when every agent arrives at a common point; this phenomenon is referred to as a *consensus*.

- To guarantee that the model operates 'well', one needs to verify the emergence of consensus.

Definition

Let $(x_n^i(\omega))_{i\in[N],\ n\geq 0}$ be a sample path, labeled by ω, of a discrete stochastic process (DCBO).

1. A sample path $(x_n^i(\omega))$ exhibits *consensus* if

$$\lim_{n\to\infty} \|x_n^i(\omega) - x_n^j(\omega)\| = 0, \quad \forall i,j \in [N].$$

2. A sample path $(x_n^i(\omega))$ exhibits *convergence* if there exists a vector $x_\infty(\omega)$ such that

$$\lim_{n\to\infty} \|x_n^i(\omega) - x_\infty(\omega)\| = 0, \quad \forall i \in [N].$$

Proposition

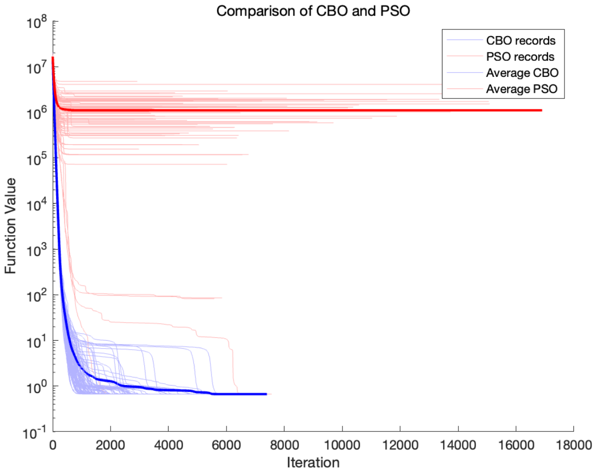
Let (x_n^i) be a iterations generated by the DCBO Algorithm. Then $f(p_n)$ is monotone decreasing in n. In particular, $p_n \in S$ for each $n \in \mathbb{N}$.

Proof. Let i_n be a random index satisfying $p_n = x_n^{i_n}$. Then for each event, $x_{n+1}^{i_n} = x_n^{i_n}$ and

$$f(p_n) = f(x_n^{i_n}) = f(x_{n+1}^{i_n}) \geq f(p_{n+1}).$$

Since $f(p_0) < \infty$, we have $f(p_n) < \infty$ for each n, and therefore $p_n \in S$.

Illustration



Theorem

Suppose that f is a Borel measurable function and parameters are well-prepared. Let (x_n^i) be the iterations generated by the DCBO Algorithm. Then, for almost every ω, either the best value $f(p_n)$ continues to update, or $(x_n^i(\omega))$ exhibits convergence.

Proof. It suffices to prove that at least on of the followings happen with probability 1:

(a) $f(p_n)$ updates its value infinitely often. That is, there exists a strictly decreasing subsequence of $f(p_n)$:

$$f(p_{t_n}) > f(p_{t_{n+1}}), \quad t_n < t_{n+1}, \quad \forall t \in \mathbb{N}, \quad \lim_{m\to\infty} t_m = \infty.$$

(b) There exists a vector $x_\infty \in S$ such that

$$\lim_{n\to\infty} \max_i \|x_n^i - x_\infty\| = 0.$$

1. Suppose that (a) does not happen. We consider the index i satisfying $F^i = F_1$.

2. For a vector $v_n^i \in \mathbb{R}^d$, let its k-th component be $v_n^{i,k}$. Then for each $k \in [d]$, one eventually has

$$x_{n+1}^{i,k} - p_\infty^k = (1 - \gamma^1 - \gamma^2 \eta_n^{i,k})(x_n^{i,k} - p_\infty^k).$$

Therefore,

$$|x_{n+1}^{i,k} - p_\infty^k| \leq |1 - \gamma^1 - \gamma^2 \eta_n^{i,k}||x_n^{i,k} - p_\infty^k|.$$

3. Then, by the strong law of large numbers, we have a desired result. □

In fact, the emergence of a consensus state is almost characterized by the convergence of the relative change in the best position.

Proposition (Consensus criterion)

Let f be a measurable function and (x_n^i) be the iterations generated by the DCBO Algorithm. Suppose that parameters are well-prepared. Then for almost every ω, a sample path $(x_i^n(\omega))$ exhibits consensus if and only if $\lim_{n\to\infty} \|p_n - p_{n+1}\| = 0$.

Proof.

- We consider the index i satisfying $F^i = F_1$.
- Obviously, consensus implies $\|p_n - p_{n+1}\| \to 0$ because for each n,

$$p_n, p_{n+1} \in \{x_{n+1}^i \mid i \in [N]\}.$$

- To prove the converse, assume that $\|p_n - p_{n+1}\| \to 0$.

- By the direct calculation,

$$x_{n+1}^{i,k} - p_{n+1}^{k} = (1 - \gamma^1 - \gamma^2 \eta_n^{i,k})(x_n^{i,k} - p_n^k) + (p_n^k - p_{n+1}^k).$$

- We take the absolute value and apply the triangle inequality to yield

$$|x_{n+1}^{i,k} - p_{n+1}^{k}| \le |1 - \gamma^1 - \gamma^2 \eta_n^{i,k}||x_n^{i,k} - p_n^k| + |p_n^k - p_{n+1}^k| =: X_n^{i,k}|x_n^{i,k} - p_n^k| + |p_n^k - p_{n+1}^k|.$$

- Then a straightforward induction leads

$$|x_n^{i,k} - p_n^k| \le |x_0^{i,k} - p_0^k| \prod_{I=0}^{n-1} X_I^{i,k} + \sum_{J=0}^{n-1} |p_J^k - p_{J+1}^k| \prod_{K=J+1}^{n-1} X_K^{i,k}, \text{ where } \prod_{K=n}^{n-1} X_K^{i,k} = 1.$$

- For each m, $\prod_{I=m}^{n-1} X_I^{i,k}$ converges to zero almost surely from the SLLN.
- **Claim.** For $Z_{n-1}^{i,k} := \sum_{J=0}^{n-1} \prod_{K=J+1}^{n-1} X_K^{i,k}$, $\lim_{n\to\infty} Z_n^{i,k}$ is finite almost surely.

- Let $\mu := \mathbb{E}[X_n^{i,k}]$. Since $Z_n^{i,k} = X_n^{i,k} Z_{n-1}^{i,k} + 1$,

$$\mathbb{E}[Z_{n+1}^{i,k} \mid \mathcal{F}_n] - \frac{1}{1-\mu} = \mu \left(Z_n^{i,k} - \frac{1}{1-\mu} \right),$$

where $\mathcal{F}_n$ is a σ-algebra generated by η_m^j for $m \in [n] \cup \{0\}$ and $j \in [N]$.

- From the martingale convergence theorem, $\lim_{n\to\infty} Z_n$ is almost surely finite.
- We have $\| p_m - p_{m+1} \| < \varepsilon$ for any $\varepsilon > 0$ whenever $m > M_\varepsilon$. This yields

$$\begin{aligned}
|x_n^{i,k} - p_n^k| \leq & |x_0^{i,k} - p_0^k| \prod_{I=0}^{n-1} X_I^{i,k} + \sum_{J=0}^{n-1} |p_J^k - p_{J+1}^k| \prod_{K=J+1}^{n-1} X_K^{i,k} \\
= & \underbrace{|x_0^{i,k} - p_0^k| \prod_{I=0}^{n-1} X_I^{i,k}}_{\to 0 \text{ a.s.}} \\
& + \underbrace{\sum_{J=0}^{M_\varepsilon} |p_J^k - p_{J+1}^k| \prod_{K=J+1}^{n-1} X_K^{i,k}}_{\to 0 \text{ a.s.}} + \underbrace{\sum_{J=M_\varepsilon+1}^{n-1} |p_J^k - p_{J+1}^k| \prod_{K=J+1}^{n-1} X_K^{i,k}}_{\leq \varepsilon Z_{n-1}}. \quad \square
\end{aligned}$$

Theorem (Boundedness implies consensus or global optimization)

Let (x_n^i) be the iterations generated by the DCBO Algorithm, where $F^j = F_2$ for each $j \in [N]$ and parameters are well-prepared. Suppose that

1. *Objective function f is continuous,*
2. *and sub-level set of p_0, $\{x \in S \mid f(x) \le f(p_0)\}$, is bounded.*

Then for almost every sample path, either $f(p_n)$ converges to $\min f$ or the agents exhibit consensus. In particular, if f has a unique global minimizer, then almost every sample path exhibit consensus.

Sketch of proof.

- **Idea.** Whenever $\|p_{n-2} - p_{n-1}\| \ge \varepsilon$, there is a chance for global optimization!
- **Observation.** Consider a sub-level set $S_\varepsilon^f = \{x \in S \mid f(x) \le \min f + \varepsilon\}$. Then there exists a positive constant D_1 such that

$$\begin{aligned}\mathbb{P}[(p_{n-1} \notin S_\varepsilon^f) \wedge (p_n \in S_\varepsilon^f)] &\ge D_1 \mathbb{P}[\bar{\Omega}_n] \\ &=: D_1 \mathbb{P}[(p_{n-1} \notin S_\varepsilon^f) \wedge (\|p_{n-2} - p_{n-1}\| \ge \varepsilon)].\end{aligned}$$

- Since $f(p_n)$ (monotonically) decreases in n,

$$\mathbb{P}[\lim_{n\to\infty} f(p_n) \le \min f + \varepsilon] \ge \sum_{n=2}^{\infty} \mathbb{P}[(f(p_{n-1}) \notin S_\varepsilon^f) \wedge (f(p_n) \in S_\varepsilon^f)] \ge D_1 \sum_{n=2}^{\infty} \mathbb{P}[\bar{\Omega}_n].$$

- Since the leftmost side is at most 1, the Borel-Cantelli lemma implies

$$\mathbb{P}[(p_{n-1} \in S_\varepsilon^f) \vee (\|p_{n-2} - p_{n-1}\| < \varepsilon) \text{ eventually}] = 1.$$

- Therefore

$$\mathbb{P}[(\lim_{n\to\infty} f(p_n) \le \min f + \varepsilon) \vee (\limsup_{n\to\infty} \|p_{n-2} - p_{n-1}\| \le \varepsilon)] = 1.$$

- Since the choice of $\varepsilon > 0$ is arbitrary, the continuity of measure leads

$$\mathbb{P}[(\lim_{n\to\infty} f(p_n) = \min f) \vee (\lim_{n\to\infty} \|p_{n-2} - p_{n-1}\| = 0)] = 1.$$

- Since the zero convergence of $\|p_{n-2} - p_{n-1}\|$ is equivalent to consensus, we have a desired result. $\square$

Global Optimization

- **Observation.** Given a measurable function f, the transition

$$y_n := (x_n^1, x_n^2, \cdots, x_n^N) \mapsto (x_{n+1}^1, x_{n+1}^2, \cdots, x_{n+1}^N) =: y_{n+1}$$

  is memoryless. Thus $y_n \mapsto y_{n+1}$ is a Markov process.

- Define $A_\varepsilon^f \subset \mathbb{R}^{Nd}$ as

$$y \in A_\varepsilon^f \quad \Leftrightarrow \quad x_i \in S_\varepsilon^f \text{ for some } i \in [N].$$

- Then, for a transition probability kernel[2] P and $\mu =: \mathrm{Law}(X_{\mathrm{in}})$,

$$\begin{aligned} &\mathbb{P}[\ x_n^i \in S_\varepsilon^f \text{ for some } i \in [N], n \in \mathbb{N}\ ] \\ &\quad = \int_{y_0 \in A_\varepsilon^f} \mu(dy_0) + \sum_{n=1}^{\infty} \int_{y_0 \notin A_\varepsilon^f} \cdots \int_{y_{n-1} \notin A_\varepsilon^f} \mu(dy_0) P(y_0, dy_1) \cdots P(dy_{n-1}, A_\varepsilon^f) \\ &\quad \geq \int_{y_0 \in A_\varepsilon^f} \mu(dy_0) = 1 - (1 - \mathrm{Law}(X_{in})(S_\varepsilon^f))^N. \end{aligned}$$

[2] P is a transition probability kernel if P is suitably measurable, each $P(\cdot, A)$ is non-negative, and each $P(x, \cdot)$ is a probability measure.

- **Example.** Consider the Ackley function

$$F_{Ack}(x) := -a\exp\left(-b\sqrt{\frac{\|x\|^2}{d}}\right) - \exp\left(\frac{1}{d}\sum_{i=1}^{d}\cos(cx_i)\right) + a + \exp(1), \qquad S = [-L, L]^d.$$

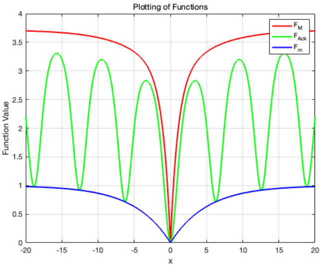
- If x_0^i is drown uniformly from S, then with a probability not lower than

$$1 - \left(1 - \frac{\pi^{d/2}(\omega_{Ack}^{-1}(\varepsilon))}{L^d\Gamma(d/2+1)}\right)^N,$$

$$\text{where}\quad \omega_{Ack}(t) = a(1 - \exp(-bt/\sqrt{d})) + \exp(1)\left(1 - \exp(-ct/\sqrt{d})\right),$$

we have $f(x_\infty) - f(x^*) < \varepsilon$ for small ε. Furthermore, in this case we have

$$\|x^* - x_\infty\| < 2\frac{\sqrt{d}}{b}\log(\frac{a}{a-\varepsilon}).$$



Proposition

Suppose f is continuous and a global minimizer exists within the support of an initial distribution.

1. Let p_∞ be an output of the DCBO algorithm. Then, for any $\varepsilon > 0$, the probability that

$$f(p_\infty) < \min f + \varepsilon \tag{1}$$

   is not less than $1-(1-P_\varepsilon)^N$, where $P_\varepsilon > 0$ is a positive constant may depending on X_{in}, dimension d, regularity of the objective function, and ε. Consequently, the probability converges to 1 as $N \to \infty$.

2. If a continuous function f_m exists such that

$$f_m(x-x^*) \le f(x) - f(x^*) \quad \text{for any global minimizer } x^* \text{ of } f,$$
$$f_m(0) = 0, \quad \text{and} \quad f_m \text{ has a strict local minimum at the origin,}$$

   then p_∞ converges to a global minimizer as $\varepsilon \searrow 0$, provided that (1) occurs.

3. Let p_m be an output of the DCBO algorithm, where p_{m-1} is included in the initial data. Then $f(p_m)$ converge to the global minimum.

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Portfolio Optimization

- **Goal.** Minimize the Sharpe ratio:
$$\text{minimize } -\frac{w^\top \mu}{w^\top \Sigma w} \text{ subject to } \{w \mid \textstyle\sum_{i=1}^d w^i = 1,\ w^i \geq 0\},$$
where μ and Σ are expectation and covariance of multi-asset.

- To get μ and Σ, we used $d = 6$ assets (AAPL, MSFT, SBUX, DB, GLD, TSLA) from 2019-01-01 to 2020-11-30.

- The original experiment used the following model with homogeneous noises:[3]
$$x_{n+1}^i = x_n^i + h\lambda(\bar{x}_n - x_n^i) + \sigma\sqrt{h}(\bar{x}_n - x_n^i)\eta_n.$$
$$\bar{x}_n = \frac{\sum_{k=1}^N x_n^i w_f^\beta(x_n^i)}{\sum_{k=1}^N w_f^\beta(x_n^i)}, \quad w_f^\beta(x) := e^{-\beta f(x)}, \quad \beta > 0.$$

[3] This corresponds to Model 2 with homogeneous noises in Figure

Results

Table: Comparison between DCBO and CBO (Model 2).

	$\beta = 10$	$\beta = 10^2$	$\beta = 10^4$	$\beta = 10^6$	DCBO
Iteration	1892.98	1925.71	1908.69	1896.54	74.29
Function value	-1.80051	-1.95519	-1.98411	-1.98450	-2.01450
Distance	0.339658	0.145540	0.109392	0.110149	0.000016

- A global minimizer is found by Sequential Least Squares Programming (SLSQP) optimizer in python. A global minimum obtained by SLSQP is -2.01450.
- For the Model 2, we used $N = 100, \lambda = 0.5, \sigma = 1, h = 10^{-2}$.
- For DCBO, we used $N = 100, \gamma^1 = 0.5, \gamma^2 = 1, \bar{\gamma}^1 = 0.4, \bar{\gamma}^2 = 0.7$.
- Each results were averaged over 100 runs.

Tests on benchmark problems

- **Test 1.** DCBO without repeat VS PSO.
- **Test 2.** DCBO with repeat VS hmPSO[4].
- For each test, we used total of 102 benchmark functions.
- Dimension of functions range from 2 to 80.
- Experiments were performed for $N = 50, 100, 200$.
- We conducted 100 tests and recorded the results.

[4] Choi, K. P., Kam, E. H. H., Tong, X. T., Wong, W. K.: *Appropriate noise addition to metaheuristic algorithms can enhance their performance.* Scientific Reports, **13**(1), 5291. (2023)

PSO and hmPSO

Initialization. Initialize x_0^i and v_0^i. Select x_0^* as the best x_0^i.

Step 1. Get i.i.d. $U_n^{i,1}$, $U_n^{i,2}$ from the uniform distribution $[0,1]^d$ and update

$$v_{n+1}^i = wv_n^i + c_1 U_n^{i,1} \odot (x_n^{i,*} - x_n^i) + c_2 U_n^{i,2} \odot (x_n^* - x_n^i).$$

Step 2. For each i, update x_n^i as $x_{n+1}^i = x_n^i + v_{n+1}^i$.

Step 2'. If using hmPSO, for $1 \le i \le n/2$,

$$x_{n+1}^i = \chi_S(\chi_S(x_n^i + v_{n+1}^i) + w_n^i), \quad \text{where} \quad \chi_S(x) = \arg\min_{y \in S} |y - x|.$$

Above, w_n^i's are i.i.d. drawn from $\mathcal{N}(0, \sigma I)$.

Step 3. For each i, if $f(x_n^{i,*}) > f(x_{n+1}^{i,*})$, then update $x_{n+1}^{i,*}$ to x_{n+1}^i.

Step 4. For each i, if $f(x_n^*) > f(\min_{j \in [N]} x_{n+1}^{j,*})$, then update x_{n+1}^* to the $x_{n+1}^{i,*}$ with the smallest function value.

Result

Table: Comparison between DCBO without repeat and PSO

# of agnets	N=50			N=100			N=200		
Statistics	**Min**	**Mean**	**Med**	**Min**	**Mean**	**Med**	**Min**	**Mean**	**Med**
CBO > PSO	19	49	34	19	42	33	18	40	32
CBO = PSO	70	22	51	75	34	55	75	39	56
CBO < PSO	13	31	17	8	26	14	9	23	14

Table: Comparison between DCBO with repeat and hmPSO.

# of agnets	N=50			N=100			N=200		
Statistics	**Min**	**Mean**	**Med**	**Min**	**Mean**	**Med**	**Min**	**Mean**	**Med**
CBO > PSO	21	41	25	17	39	25	17	38	26
CBO = PSO	74	52	67	80	57	70	83	60	72
CBO < PSO	7	9	10	5	6	12	7	4	4

Table: Comparison for high dimensional ($d = 80$) functions up to 4 significant digits

	DCBO without repeat			**PSO**		
	$N = 50$	$N = 100$	$N = 200$	$N = 50$	$N = 100$	$N = 200$
Ackley (Global min = 0)						
Min	0	0	0	4.911	2.351	1.155
Mean	4.504	2.145	1.064	15.73	13.57	11.35
Median	0	0	0	16.03	14.55	13.02
Mean iter	3577	2983	2699	2875	2146	1733
Rastrigin (Global min = 0)						
Min	201.0	84.57	25.87	378.3	354.4	317.5
Mean	351.2	211.6	93.40	551.8	511.0	486.9
Median	345.7	205.5	91.04	544.7	512.8	494.9
Mean iter	2988	2994	2893	2789	2010	1573
Trid (Global min = -88480)						
Min	-88460	-88480	-88480	5125000	9860000	2551000
Mean	-72790	-78980	-84100	60300000	67510000	63390000
Median	-75250	-81760	-85200	54040000	64280000	54150000
Mean iter	40000	40000	40000	22210	18430	18890
Rosenbrock (Global min = 0)						
Min	3.998	0.03853	0.008562	166600	224900	55520
Mean	52.72	44.00	36.18	689200	645100	620400
Median	50.29	43.92	36.65	669000	637900	626000
Mean iter	40000	40000	40000	12250	8827	13140
Powell (Global min = 0)						
Min	0.000017	0.000005	0.000001	551.4	716.9	482.6
Mean	0.000023	0.000006	0.000002	8958	8724	7852
Median	0.000024	0.000006	0.000002	8210	8472	7847
Mean iter	40000	40000	40000	19340	18680	18570
Styblinski-Tang (Global min = -3133)						
Min	-2921	-3006	-3105	-2794	-2851	-2836
Mean	-2795	-2882	-2989	-2661	-2663	-2681
Median	-2794	-2879	-2992	-2667	-2667	-2681
Mean iter	2509	2384	2289	3151	2042	1547

Table: Comparison for high dimensional ($d = 80$) functions with 40,000 iterations up to 4 significant digits

	DCBO with repeat			hmPSO		
	$N = 50$	$N = 100$	$N = 200$	$N = 50$	$N = 100$	$N = 200$
Ackley (Global min = 0)						
Min	0	0	0	1.646	1.228	0
Mean	3.332	1.246	0.1691	12.34	6.639	0.9192
Median	0	0	0	15.09	3.001	0
Rastrigin (Global min = 0)						
Min	35.82	1.990	0	390.2	366.1	304.6
Mean	149.9	51.63	10.80	545.6	514.5	511.2
Median	146.8	55.22	2.985	552.5	513.7	505.2
Trid (Global min = -88480)						
Min	-88270	-88460	-88480	-87970	-88060	-86970
Mean	-71140	-79580	-83290	55670	72270	62890
Median	-74700	-81020	-84360	-16570	-7398	-15270
Rosenbrock (Global min = 0)						
Min	17.63	0.01050	0.009896	0.000037	0	0
Mean	55.79	49.62	34.57	135.4	96.15	98.46
Median	50.52	44.10	36.65	28.46	15.19	14.77
Powell (Global min = 0)						
Min	0.000019	0.000004	0.000001	118.8	43.65	75.18
Mean	0.000025	0.000006	0.000002	352.1	394.0	413.3
Median	0.000024	0.000006	0.000002	325.4	393.0	418.1
Styblinski-Tang (Global min = -3133)						
Min	-3133	-3133	-3133	-2794	-2879	-2836
Mean	-3064	-3128	-3133	-2674	-2692	-2686
Median	-3063	-3133	-3133	-2681	-2695	-2681

- Each algorithms generally performed well in global optimization.
- PSO < CBO without repeat < hmPSO < CBO with repeat
- For functions where optimization is well-performed, the algorithm terminates with fewer iterations as the number of agents increases:

	CBO without repeat			**PSO**		
	$N = 50$	$N = 100$	$N = 200$	$N = 50$	$N = 100$	$N = 200$
Ackley (Global min = 0)						
Min	0	0	0	4.911	2.351	1.155
Mean	4.504	2.145	1.064	15.73	13.57	11.35
Median	0	0	0	16.03	14.55	13.02
Mean iter	3577	2983	2699	2875	2146	1733
Griewank (Global min = 0)						
Min	0	0	0	0.03687	0	0
Mean	0.004187	0.004656	0.003203	201.5	182.4	153.5
Median	0	0	0	180.9	180.7	180.1
Mean iter	3617	3160	2819	3230	2141	1627
Zakharov (Global min = 0)						
Min	0	0	0	104.3	25.03	0
Mean	0	0	0	649.0	446.1	285.3
Median	0	0	0	602.0	379.8	257.6
Mean iter	7452	5626	4578	34770	31760	29270

- This likely stems from the tendency that the occurrence of p_n such that $f(p_n) \cong \min f$ is faster for larger values of N.

Compressed Sensing

Let $Ax = b$, where x is sparse.

Goal. Given A and b, find x.

$$\begin{bmatrix}0.89\\0.18\\0.69\end{bmatrix} = \begin{bmatrix}1&0&0&0&0&0&0&0\\0&1&0&0&0&0&0&0\\0&0&0&0&1&0&0&0\end{bmatrix}\begin{bmatrix}0.89\\0.18\\0.03\\0.37\\0.69\\0.54\\0.11\\0.90\end{bmatrix} \qquad \begin{bmatrix}0\\0.18\\0\end{bmatrix} = \begin{bmatrix}1&0&0&0&0&0&0&0\\0&1&0&0&0&0&0&0\\0&0&0&0&1&0&0&0\end{bmatrix}\begin{bmatrix}0\\0.18\\0.03\\0\\0\\0\\0.11\\0\end{bmatrix}$$

$$\begin{bmatrix}0.2336\\-0.1767\\-0.1572\end{bmatrix} = \begin{bmatrix}0.52&0.67&0.65&0.53&-2.30&-0.89&0.85&-0.11\\0.21&-0.84&-1.62&-2.06&-1.16&-0.02&0.21&0.93\\-1.34&-1.38&-0.37&-0.22&2.09&-0.62&0.93&-1.40\end{bmatrix}\begin{bmatrix}0\\0.18\\0.03\\0\\0\\0\\0.11\\0\end{bmatrix}$$

Figure: Obtaining 3 output values in different manners.[5]

[5] https://www.codeproject.com/Articles/852910/Compressed-Sensing-Intro-Tutorial-w-Matlab

- This problem might be tackled by solving

$$\text{Minimize } \|Ax - b\|_2^2 \text{ subject to } \|x\|_0 < r.$$

- We consider a relaxed problem

$$\text{Minimize } \|Ax - b\|_2^2 \text{ subject to } \|x\|^p < r.$$

- More precisely, we apply DCBO to minimize f, where

$$f(x) = \begin{cases} \frac{1}{2}\|Ax - b\|^2, & \text{if } \|x\|_{0.5} \le r, \\ \infty, & \text{if } \|x\|_{0.5} > r, \end{cases} \quad p = 0.5.$$

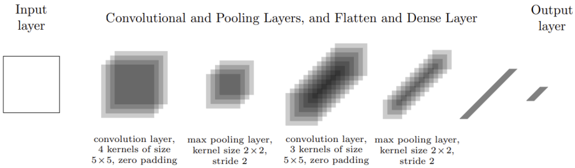
  Note that the effective domain is *non-convex*.

- After finding x, we applied hard-thresholding and minimized $\|Ax - b\|_2^2$, while maintaining the sparsity.

Results ($d = 100$)

PGD					
Radius (r)	6	12	18	24	30
s=2 (TPR)	0.9200(0.0222)	0.2950(0.0334)	0.0850(0.0202)	0.0300(0.0119)	0.0200(0.0098)
s=2 (FPR)	0.0051(0.0010)	0.0313(0.0013)	0.0350(0.0010)	0.0363(0.0009)	0.0365(0.0008)
s=4 (TPR)	0.5000(0.0036)	0.8950(0.0225)	0.3750(0.0317)	0.0850(0.0164)	0.0475(0.0122)
s=4 (FPR)	0.0002(0.0002)	0.0049(0.0010)	0.0285(0.0010)	0.0340(0.0010)	0.0355(0.0011)
s=6 (TPR)	0.3367(0.0058)	0.5950(0.0163)	0.4667(0.0331)	0.1817(0.0260)	0.0967(0.0142)
s=6 (FPR)	0.0005(0.0003)	0.0030(0.0008)	0.0170(0.0015)	0.0311(0.0010)	0.0340(0.0011)
DCBO ($N = 200$)					
Radius (r)	6	12	18	24	30
s=2 (TPR)	0.8550(0.0278)	0.9200(0.0184)	0.9600(0.0169)	0.9800(0.0121)	0.9500(0.0167)
s=2 (FPR)	0.0093(0.0017)	0.0130(0.0027)	0.0077(0.0023)	0.0200(0.0043)	0.0388(0.0059)
s=4 (TPR)	0.4275(0.0152)	0.6625(0.0278)	0.8700(0.0218)	0.8575(0.0217)	0.8500(0.0259)
s=4 (FPR)	0.0093(0.0012)	0.0206(0.0017)	0.0163(0.0023)	0.0265(0.0033)	0.0333(0.0040)
s=6 (TPR)	0.2933(0.0138)	0.4250(0.0171)	0.5733(0.0219)	0.7383(0.0259)	0.7683(0.0227)
s=6 (FPR)	0.0116(0.0014)	0.0184(0.0015)	0.0235(0.0019)	0.0256(0.0023)	0.0351(0.0030)
DCBO ($N = 2000$)					
Radius (r)	6	12	18	24	30
s=2 (TPR)	0.9350(0.0197)	0.9750(0.0131)	0.9700(0.0119)	1.0000(0.0000)	0.9900(0.0070)
s=2 (FPR)	0.0040(0.0012)	0.0029(0.0014)	0.0055(0.0022)	0.0030(0.0013)	0.0100(0.0032)
s=4 (TPR)	0.4950(0.0167)	0.7925(0.0268)	0.9050(0.0221)	0.9400(0.0178)	0.9750(0.0104)
s=4 (FPR)	0.0055(0.0010)	0.0130(0.0016)	0.0095(0.0020)	0.0074(0.0020)	0.0060(0.0022)
s=6 (TPR)	0.3383(0.0126)	0.5050(0.0168)	0.6683(0.0209)	0.8467(0.0235)	0.9100(0.0210)
s=6 (FPR)	0.0073(0.0010)	0.0119(0.0013)	0.0140(0.0017)	0.0145(0.0022)	0.0116(0.0025)

MNIST via CBO



(b) CNN (LeNet-1), cf. [46, Section III.C.7], with two convolutional and two pooling layers, and one dense layer

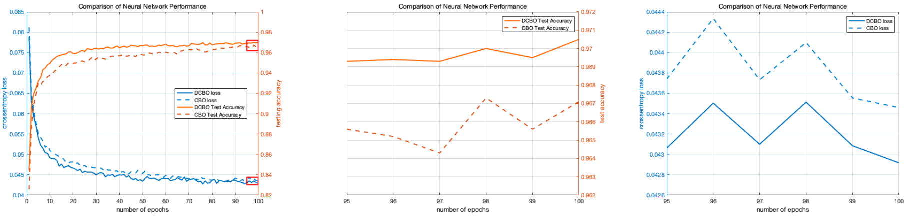
- We compared CBO and DCBO.
- **Goal.** For the forward pass f_θ and training samples $(x^j, y^j)_{j=1}^M$

  $$\text{minimize } \tfrac{1}{M} \textstyle\sum_{j=1}^M \ell(f_\theta(x^j), y^j),$$

  where the loss function is the categorical crossentropy loss:

  $$\ell(\hat{y}, y) = -\sum_{k=0}^{9} y_k \log(\hat{y}_k) \quad \hat{y} = f_\theta(x).$$

- 2112 parameters, $N = 100$ agents, batch size $n_\mathcal{E} = 60$ and $n_N = 10$.



(a) DCBO vs CBO (b) Test accuracy (c) Crossentropy loss

Table: Comparison between DCBO and CBO.

	DCBO	CBO
Test accuracy	0.9705	0.9671
Objective value	0.0429	0.0435

Outline

1. Introduction
2. Emergence of Consensus and Global Optimization
3. Numerical Simulations
4. Conclusion

- We present DCBO algorithm, which is motivated by consensus-based optimization methods.

- DCBO is distinguished by being discrete but not a discretized CBO, yet operating as a CBO.

- Verifying the emergence of consensus under the heterogeneous noises was an open problem at the particle level, and we partially address this problem.

- DCBO can be applied to general optimization problems with reasonable results.

The End